



# Setting Up Docker & n8n

## Step-by-Step Guide for Students

Works on Mac & Windows • No prior experience needed

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### What Is This Guide?

This guide will help you install Docker Desktop and run n8n on your laptop. Once complete, you will be able to build and run AI automation agents in the classroom.

*You do not need any technical background. Just follow each step in order and you will be ready in under 15 minutes.*

### What You Will Install

| Tool           | What It Does                                       | Cost                 |
|----------------|----------------------------------------------------|----------------------|
| Docker Desktop | Runs applications in containers on your laptop     | ✓ Free               |
| n8n            | The automation tool where you build your AI agents | ✓ Free (self-hosted) |

 **Time needed:** About 10–15 minutes total. Follow each step carefully and you will be done before class starts!

## PART 1

### Installing Docker Desktop

Docker is a tool that runs software in isolated "containers". Think of a container like a sealed box that contains everything n8n needs to run — no messy installation, no conflicts with other software on your laptop.

## Step 1 — Go to the Docker website

Open your browser and go to:

<https://www.docker.com/products/docker-desktop>

## Step 2 — Download for your operating system

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                           |
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|  <b>Mac Instructions</b><br>Click "Download for Mac"<br>If you have an M1/M2/M3 chip (Apple Silicon) → choose "Mac with Apple Chip"<br>If you have an older Mac (Intel) → choose "Mac with Intel Chip"<br><br><b>Not sure which chip you have?</b><br>Click the  Apple menu → "About This Mac" → look for "Chip" or "Processor" |  <b>Windows Instructions</b><br>Click "Download for Windows"<br>Download the installer (.exe file)<br><br><b>Requirements:</b> <ul style="list-style-type: none"><li>• Windows 10 or 11 (64-bit)</li><li>• WSL 2 must be enabled (Docker will prompt you to do this if needed)</li></ul> |
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## Step 3 — Install Docker Desktop

|                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                      |
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|  <b>Mac Instructions</b> <ol style="list-style-type: none"><li>1. Open the downloaded .dmg file</li><li>2. Drag the Docker icon into your Applications folder</li><li>3. Open Docker from Applications or Spotlight (Cmd + Space → type "Docker")</li><li>4. macOS may ask for your password — enter it and click OK</li></ol> |  <b>Windows Instructions</b> <ol style="list-style-type: none"><li>1. Double-click the downloaded .exe installer</li><li>2. Follow the installation wizard — click Next and accept defaults</li><li>3. Restart your computer if prompted</li><li>4. Docker Desktop launches automatically after restart</li></ol> |
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## Step 4 — Wait for Docker to start

When Docker Desktop opens, wait until you see the status at the bottom left change to:

 **Ready signal:** "Engine running" shown in green at the bottom left of the Docker Desktop window. This means Docker is ready!

 **First launch:** The first time Docker starts it may take 1–2 minutes. Do not close the window. Wait for the green "Engine running" status.

## Step 5 — Verify Docker is working

Open Terminal (Mac) or Command Prompt (Windows) and type:

```
docker --version
```

You should see something like: **Docker version 27.x.x**

 **How to open Terminal:** Mac: Press Cmd + Space, type "Terminal", press Enter | Windows: Press Windows key, type "cmd", press Enter

## Installing and Running n8n

n8n is the automation platform where you will build your AI agents. We run it inside Docker using a single command. Your workflows are saved permanently even if you restart your computer.

### Step 1 — Open Terminal or Command Prompt



#### Mac Instructions

Press Cmd + Space to open Spotlight  
Type: Terminal  
Press Enter



#### Windows Instructions

Press Windows key + R  
Type: cmd  
Press Enter

### Step 2 — Run n8n using Docker

Copy and paste this command into your Terminal / Command Prompt and press Enter:

#### Mac command:

```
docker run -d \ --name n8n \ --restart unless-stopped \ -p 5678:5678 \ -v n8n_data:/home/node/.n8n \ docker.n8n.io/n8nio/n8n
```

#### Windows command (Command Prompt):

```
docker run -d ^ --name n8n ^ --restart unless-stopped ^ -p 5678:5678 ^ -v n8n_data:/home/node/.n8n ^ docker.n8n.io/n8nio/n8n
```

#### What does each part mean?

|                                       |                                                      |
|---------------------------------------|------------------------------------------------------|
| <code>-d</code>                       | Runs in background — you can close Terminal after    |
| <code>--name n8n</code>               | Gives the container a name so you can find it easily |
| <code>--restart unless-stopped</code> | Auto-restarts n8n after every computer reboot        |

|                           |                                                          |
|---------------------------|----------------------------------------------------------|
| <code>-p 5678:5678</code> | Makes n8n accessible at localhost:5678 in your browser   |
| <code>-v n8n_data</code>  | Saves your workflows permanently — they survive restarts |

 **First time:** Docker will download n8n automatically (about 500MB). This takes 1–3 minutes depending on your internet speed. You will see text scrolling — this is normal. Wait until it stops.

### Step 3 — Wait for n8n to start

After the command finishes, wait about 20–30 seconds for n8n to initialise.

### Step 4 — Open n8n in your browser

Open any browser (Chrome, Firefox, Edge, Safari) and go to:

**`http://localhost:5678`**

### Step 5 — Create your local account

n8n will ask you to create an account. Fill in:

- Any name (e.g. your real name or a nickname)
- Any email address (does not need to be real)
- Any password (just remember it!)

 **Privacy note:** This account is stored only on your laptop. Nothing is sent to the internet. n8n never asks for payment or personal details.

 **Success signal:** You should see the n8n "Overview" dashboard with tabs for Workflows, Credentials, Executions, Variables, and Data tables. n8n is ready!

## Verifying Everything Works

Let's make sure everything is set up correctly before class.

| # | Check                                               | Expected Result                                                                                                      |
|---|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 1 | Docker Desktop is open and running                  | "Engine running" in green       |
| 2 | <code>http://localhost:5678</code> loads in browser | n8n login or dashboard appears  |
| 3 | You can log in to n8n                               | Overview dashboard is visible   |

|   |                                         |                                    |
|---|-----------------------------------------|------------------------------------|
| 4 | n8n container visible in Docker Desktop | "n8n" container shown as running ✓ |
|---|-----------------------------------------|------------------------------------|



## After Restarting Your Computer

Because we used `--restart unless-stopped`, n8n will restart automatically every time your computer reboots — as long as Docker Desktop is running.

Here is what to do after a reboot:

- Open Docker Desktop and wait for "Engine running"
- Open your browser and go to <http://localhost:5678>
- Your workflows and credentials will all still be there!



**Important:** Make sure Docker Desktop is set to launch automatically when you log in. Go to Docker Desktop → Settings → General → ✓ "Start Docker Desktop when you sign in to your computer"



## Troubleshooting

| Problem                          | Likely Cause                  | Fix                                                                                                     |
|----------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------|
| localhost:5678 not loading       | Container has no port mapping | Stop container, re-run the docker run command with <code>-p 5678:5678</code>                            |
| Docker says "Engine not running" | Docker Desktop not started    | Open Docker Desktop and wait for green status                                                           |
| Command not found: docker        | Docker not installed properly | Re-install Docker Desktop from <a href="https://docs.docker.com/docker-for-mac/install/">docker.com</a> |
| "name already in use" error      | Old container exists          | Run: <code>docker rm n8n</code> then try again                                                          |
| n8n loads but workflows missing  | Wrong volume / new container  | Make sure <code>-v n8n_data</code> is in your command                                                   |
| Windows: command not working     | Using wrong line continuation | Use <code>^</code> instead of <code>\</code> for line breaks on Windows cmd                             |



## Quick Reference Card

Keep this handy during the classroom session:

|                          |                                                              |
|--------------------------|--------------------------------------------------------------|
| n8n URL                  | <a href="http://localhost:5678">http://localhost:5678</a>    |
| Start n8n (if stopped)   | <code>docker start n8n</code>                                |
| Stop n8n                 | <code>docker stop n8n</code>                                 |
| Check if n8n is running  | <code>docker ps</code>                                       |
| View n8n logs            | <code>docker logs n8n</code>                                 |
| Remove and reinstall n8n | <code>docker rm n8n</code> (then run the full command again) |

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**Ready to build your first AI agent? **

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